ANUJ KALE

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SUMMARY

Mechanical Engineering graduate with experience in autonomous UAVs, ROS2 robotics, computer vision, and embedded systems. Electronics Lead for Team Squadrone's SAE AeroTHON 2025 UAV project (AIR 11) and co-author of a published UAV research paper presented at DAUS 2026. Interested in navigation, guidance, control systems, and autonomous robotics.

EDUCATION

Bachelor of Engineering (BE) in Mechanical Engineering **July 2022 - July 2026**

Pune Vidyarthi Griha's College of Engineering and Technology -Savitribai Phule Pune University

- CGPA : 8.12
- HSC : 77%
- SSC : 90.4%

Honors in Robotics & Automation

SKILLS

- **UAV & Robotics:** ROS2, Gazebo, RViz, PX4, ArduPilot, Mission Planner, UAV System Architecture
- **Programming:** Python, OpenCV, YOLO, Embedded C, Basic C++
- **Navigation & Perception:** Sensor Integration, Computer Vision, Path Planning Fundamentals
- **Embedded Systems:** Arduino, ESP32, Raspberry Pi, Microcontroller Interfacing
- **Industrial Automation:** Mitsubishi FX3G PLC, Ladder Logic
- **Mechanical & Prototyping:** SolidWorks, 3D Printing

PROJECTS

1. **Team Squadrone - Autonomous UAV for Disaster Response**

SAE AeroTHON 2025 Phase I – AIR 11 | Research Paper Published at DAUS 2026, Salzburg, Austria

- Led electronics architecture and subsystem integration for an autonomous disaster-response UAV.
- Coordinated integration of flight controller, onboard computing, communication systems, sensors and mission payloads.
- Participated in UAV component selection, sensor integration, power system planning and simulation-based validation.
- Contributed to published research on terrain-aware target detection and autonomous payload delivery for UAV applications.

2. **Autonomous Mobile Assistant Robot (ROS2)**

- Developed a ROS2-based autonomous mobile robot using mecanum-drive kinematics.
- Implemented IMU and odometry integration for localization and navigation.
- Designed a navigation pipeline for autonomous indoor mobility.
- Utilized Gazebo and RViz for simulation, visualization and testing.

3. **Quadruped Robot**

- Led development of a 12-servo quadruped robot using Arduino Nano and 3D-printed components.
- Implemented gait control algorithms and servo coordination for stable locomotion.
- Conducted hardware integration, testing and performance validation.

4. **PLC-Based Pneumatic Gripper System**

- Developed a PLC-controlled pneumatic gripping system using Mitsubishi FX3G and a 5/3 solenoid valve.
- Programmed ladder logic for forward-reverse actuation and sensor-based control.

RESEARCH & PUBLICATIONS

Autonomous UAVs for Disaster Response: Terrain-Aware Target Detection and Payload Delivery

- Published research paper at the 2nd International Conference on Drones and Unmanned Systems (DAUS 2026), Salzburg, Austria.
- Investigated autonomous UAV systems for disaster-response missions, terrain-aware target detection and precision payload delivery.
- Explored integration of computer vision, onboard sensing and autonomous mission execution concepts.

JOB SIMULATIONS & CERTIFICATIONS

Robotics & Controls Job Simulation | Forage

- Used Python-based response-time analysis to identify a slow joint operation and considered optimization, sensor alignment and validation.

Amazon Robotics: Black-Friday Fleet Coordination

- Applied A* and cooperative reservation-table planning, including vertex and edge conflict prevention, deadlock handling and fleet-size trade-off analysis. Educational simulation, not Amazon employment.

Machine Learning for Automotive Safety

- Evaluation report: 94/100. Analysed detection metrics, vulnerable-road-user false negatives, night/occlusion failure modes and ISO 21448 / SOTIF mitigations. Educational simulation.

ROS2 Udemy Course

- Learned the foundations of ROS 2 for developing modular robotic applications through practical robotics concepts, developing simulation-based workflows.

EXPERIENCE

Procurement Intern, Unbox Robotics, Pune **Dec 2024 - Jan 2025**

- Supported technical procurement activities for warehouse robotics systems using SAP.
- Coordinated component sourcing, vendor communication and engineering support operations.
- Assisted in inventory tracking, documentation management and material planning.
- Gained exposure to robotics product-development workflows and cross-functional engineering operations.

ADDITIONAL INFORMATION

- **Languages :** English, Hindi, Marathi.
- **Extracurricular Interests :** Media Head (MESA & Samsara), EDC and TEDx design/media teams.